



Aesthetic Wing

Real chill street aviation

Model: AC-AWG-01

Class: personal electric mobility / bounded lift

Status: speculative research concept

Phase I: unmanned only

The short version

Aesthetic Wing is an electric one-wheel that can make an **air step**. Most of the time it rolls. When the surface changes, two guarded lift petals open, progressively unload the wheel, rise through a bounded envelope, and settle back onto the street. It is not a flying car. It proposes a smaller gesture between riding and flying: **a little more street, in every direction.**

The research question

Can a compact one-wheel vehicle transition safely between rolling, partial lift, and controlled low hover while preserving an intuitive board-riding experience?

The interface

lean to move

press to rise

relax to settle

The rider expresses intent through stance and foot pressure. Stabilization, height limiting, energy reserve, and automatic settling remain machine duties.

Truth before spectacle

The illustration is a product vision, not flight-test evidence. Phase I contains no rider, no public-street operation, and no unrestricted altitude.



Concept illustration: Jeffrey in a controlled eight-foot hover. The vehicle depicted does not exist.



Aesthetic Wing

Phase I research program

Operating modes

Roll	One central wheel supports ordinary efficient travel.
Skim	Lift partially unloads the wheel over rough ground.
Air step	A short controlled rise, translation, and landing.
Hover	Fully airborne movement inside a deliberately low envelope.
Safe settle	Automatic descent after lost input, low reserve, or fault.

Phase I objectives

1. Model mass, lift area, power, thermal behavior, downwash, noise, and transition dynamics.
2. Validate guarded propulsion on a stationary full-load test article.
3. Develop roll-to-hover controls and automatic settling in simulation and hardware.
4. Demonstrate an unmanned weighted prototype rolling, unloading, crossing a controlled interruption, landing, and rolling again.

Safety gates

Rotor containment; independent shutdown; tethered initial trials; battery isolation; thermal monitoring; motor-out testing; bounded height and speed; formal hazard review before every expansion of the envelope.

Twelve months / \$300,000

1-2	requirements, prior art, hazards
2-4	energy and geometry trade study
3-6	guarded-lift test rig
5-8	rolling mule and controller
7-10	integrated tethered tests
9-11	unmanned air-step demonstration
11-12	validation and Phase II design



Roll → air step. Generated concept sequence.
Performance remains to be measured.

Funding request

\$300,000 over 12 months to establish technical feasibility and demonstrate an unmanned wheel-to-hover transition. The non-confidential funding thesis is a deployable distributed-lift and transition-control system for bounded personal mobility. The product and cultural identity remain **Aesthetic Wing**.